

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Annexure A

18
20
BL 1
Total Marks:30

Constraint-Based Problem Solving

Code of Conduct:

Shonima (192511387)

I Betty D. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Rural Branch Office

Problem Analysis :

The branch office has :

- No high-speed wired Internet
- Need for cloud applications and video conferencing
- Requirement for secure communication with headquarters
- Limited connectivity budget
- Need for reliable and uninterrupted operation.
- Future scalability requirement.

Recommended Solution:

Internet connection:

Use a 4G / 5G wireless broadband connection as the primary internet connection. If available, use a satellite internet connection as backup.

Networking Devices:

- 4G / 5G router
- Gigabit Ethernet switch
- Wireless access point
- Firewall / router
- VPN gateway

Security Mechanisms:

- Site - to - site VPN between branch and headquarters.
- Firewall with access - control rules.
- WPA3 for Wi-Fi security
- Strong passwords and user authentication
- Regular firmware and security updates.

Conclusion:

A 4G / 5G - based network with firewall, VPN, switch and secure Wi-Fi, supported by backup connection, provides cost effective and scalable solution for rural branch.

University Wireless Network

The university requires campus-wide Wi-Fi classrooms, laboratories, libraries and hostels while supporting many simultaneous users. The main constraints are limited APs, interference, security, coverage and cost.

Access Point Placement

Access point should be strategically installed in:

- classroom
- Laboratories
- Libraries
- Hostels
- common areas

Frequency selection:

- 5 GHz / 6 GHz
- 2.4 GHz

Interference Reduction:

- Avoid overlapping channels.
- Use automatic channel selection where supported.
- Reduce unnecessary AP transmission power.
- Place APs away from thick walls and electrical interference.
- Use directional planning in high-density areas.

Authentic and security:

WPA3-Enterprise
802.1X authentication
RADIUS server
individual student credentials
separate student, staff and
guest networks using VLANs.

Conclusion:

A well planned wireless network ensures coverage, performance, security, and low cost.

5. Financial Institution

network segmentation:

- Divide the network into VLANs for employees, database, management, and guests.
- Keep sensitive customer data in a separate, restricted VLAN.

FIREWALL:

- Use an external firewall between the internet and network
- Use an internal firewall to protect sensitive systems.
- Allow only necessary traffic.

Authentication:

- Use MFA and strong passwords
- Implement role-based access control (RBAC).
- Use 802.1X for secure network access.

Intrusion Detection:

- IDS/IPS at the network perimeter.
- continuous traffic monitoring
- security logs and alerts.
- Regular vulnerability assessment

Performance and cost optimization:

- centrally manage security device
- monitor traffic to identify bottlenecks
- prioritize critical banking applications.
- Apply VLANs instead of physically separate networks.

Conclusion:

A segment network using VLANs, DMZ, firewalls, MFA, RBAC and IDS/IPS provides strong protection for sensitive financial information without significantly reducing network performance or unnecessarily increasing operational costs.